

Differentiation

Week 3D

Overview

Further Applications of Derivatives

Key Ideas - Equation of a Tangent

Derivatives of other Functions

Key Ideas - Derivative of Exponential Functions

Key Idea – Further Applications Equation of a Tangent

Derivatives have many applications such as curve sketching, rates of change, related rates, motion in physics, optimisation problems but in this subject, we are only studying **equation of a tangent**.

Before we find equations of tangents of algebraic functions, we need to review the equation of a straight line.

The equation of a straight line may be found if we are given the slope of the line (m) and one point on the line (x_1, y_1) .

We use the point-gradient formula $y - y_1 = m(x - x_1)$ to find the equation of the line.

The equation of a straight line may be written in:

- Gradient intercept form $y = mx + b$
- General form: $ax + by + c = 0$

Steps for Finding the Equation of a Tangent

The gradient function, $f'(x)$, gives the slope of the tangent at any point x on the curve $y = f(x)$.

The gradient m at a given point (x_1, y_1) is given by $m = f'(x_1)$.

Steps to find the equation of a tangent:

1. Identify the point where the tangent touches the curve (x_1, y_1)

Use $f(x_1)$ to find y_1

(Sometimes (x_1, y_1) is given and this step is not necessary)

2. Find the slope of the tangent (m)

Differentiate $f(x)$ to find $f'(x)$ then evaluate $f'(x_1)$

3. Find the equation.

Substitute (m) and point (x_1, y_1) , into the formula $y - y_1 = m(x - x_1)$

The equation may be written in either gradient-intercept form or general form.

Example:

Find the equation of the tangent to the curve $f(x) = x^3 - 3x + 5$ at the point on the curve where $x = 2$. Write the equation in the general form.

Solution:

Step 1 **Find the y-value.**

$$f(x) = x^3 - 3x + 5$$

$$f(2) = 8 - 6 + 5 = 7. \text{ Therefore the point is } (2, 7).$$

Step 2 **Find the gradient of the tangent**

$$f(x) = x^3 - 3x + 5$$

$$f'(x) = 3x^2 - 3$$

$$f'(2) = 12 - 3 = 9, \therefore m = 9$$

Step 3 **Find the equation using $y - y_1 = m(x - x_1)$**

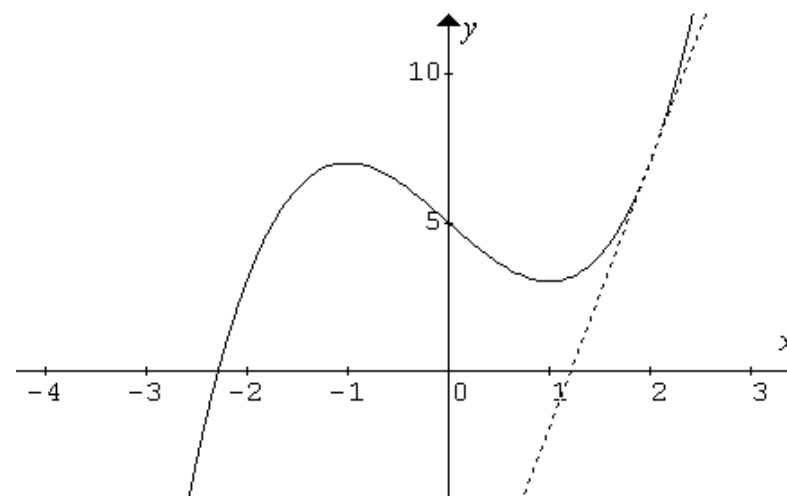
$$y - 7 = 9(x - 2)$$

$$y - 7 = 9x - 18$$

$$y = 9x - 11$$

The general form is $ax + by + c = 0$

Therefore, the general form is $9x - y - 11 = 0$.



Exercise:

Find the equation of the tangent to the curve $y = x^4 - 4x + 2$ at the point on the curve where $x = -1$. Write the equation in the general form.

1. y-coord when $x = -1$

$$y = (-1)^4 - 4 \times (-1) + 2 = 7$$

2. find the gradient = m

$$m = y' = 4x^3 - 4$$

$$\text{at } x = -1 : y'(-1) = 4 \times (-1)^3 - 4$$

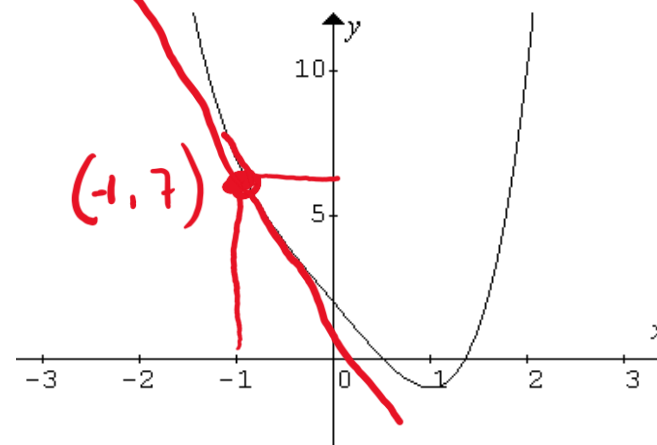
3. equation of tangent $m = -8$

$$y - y_1 = m(x - x_1)$$

$$y - 7 = -8(x - (-1))$$

$$y - 7 = -8x - 8$$

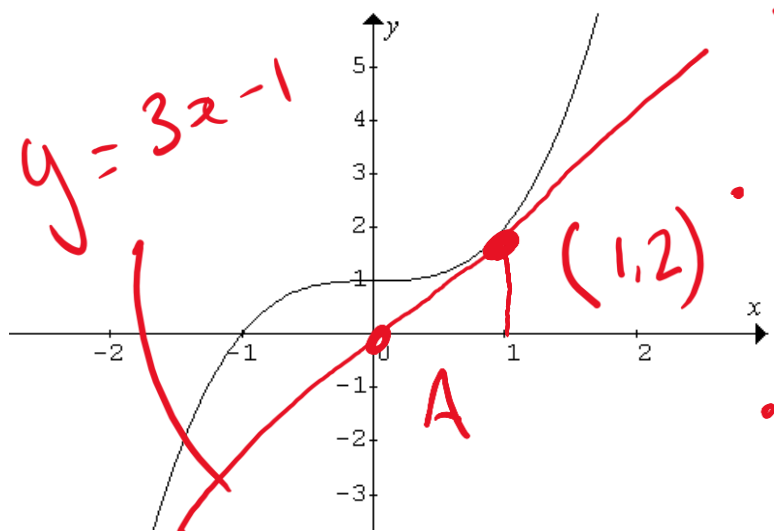
$$8x + y + 1 = 0$$



$$(x_1, y_1) = (-1, 7)$$
$$m = -8$$

Exercise:

The tangent to the curve $y = x^3 + 1$ at $x = 1$ intersects the x -axis at A . Find the coordinates of the point A .



• y -coord at $x = 1$

$$y = 1^3 + 1 = 2$$

• find the gradient at $x = 1$

$$y' = 3x^2$$

$$m = y'(1) = 3 \times 1^2 = 3$$

• equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 1$$

now to find A let $y = 0$:

$$0 = 3x - 1$$

$$3x = 1$$

$$x = \frac{1}{2}$$

$$\therefore \left(\frac{1}{2}, 0\right)$$

Exercise:

A tangent to the curve $y = \frac{1}{2}x^4$ has a gradient of 16. Find the equation of this tangent write the equation in the general form.

- Find the gradient function of $y = \frac{1}{2}x^4$

$$y' = \frac{1}{2} \times 4x^3 = 2x^3$$

- now $y' = \text{gradient of } 16$

$$2x^3 = 16$$

$$x^3 = 8$$

$$x = \sqrt[3]{8} = 2$$

$\therefore y$ -coord

$$y = \frac{1}{2} \times 2^4 = 8$$

- equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - 8 = 16(x - 2) \Rightarrow 16x - y - 24 = 0$$

Exercise:

Find the equation of the tangent to the curve $y = x^2 - 4x$ which is parallel to the line $2y = 4x - 5$.

- ^{Find} gradient which is parallel to $2y = 4x - 5$

$$y = \frac{4x}{2} - \frac{5}{2} \Rightarrow y = 2x - \frac{5}{2} \therefore \underline{m=2}$$

- now find x by setting $y = 2$

$$y' = 2x - 4 \quad \therefore \quad 2x - 4 = 2$$

$$2x = 6 \Rightarrow x = 3$$

- y -coord for $x = 3$

$$y = 3^2 - 4 \times 3 = -3$$

- equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - (-3) = 2(x - 3)$$

$$y + 3 = 2x - 6$$

$$2x + y + 9 = 0$$

Key Idea – *Derivative of Exponential Function*

When differentiating exponential functions, we still apply the same five differentiation rules we've already learned. The only difference is how the exponential function itself behaves when we differentiate it.

The derivative of the function $y = e^x$ is itself so we have,

$$\frac{dy}{dx} = e^x$$

The chain rule allows us to differentiate,

$$y = f(\quad) = e^{(\quad)}, \text{ where } (\quad) \text{ is any differentiable function.}$$

Which gives us,

$$\text{If } y = e^{f(x)} \text{ then } \frac{dy}{dx} = f'(x)e^{f(x)}$$

Remember:

“Differentiate the outside function first leaving the inside the same, then multiply by the derivative of what's inside.”

Examples:

Differentiate the following functions:

$$\text{If } y = e^{2x}, \text{ then } \frac{dy}{dx} = 2e^{2x}$$

$$\text{If } y = e^{3x}, \text{ then } \frac{dy}{dx} = 3e^{3x}$$

$$\text{If } y = e^{x^2 - 4}, \text{ then } \frac{dy}{dx} = 2xe^{x^2}$$

$$\text{If } y = e^{2x^3}, \text{ then } \frac{dy}{dx} = 6x^2 e^{2x^3}$$

$$\text{If } y = e^{3x^4 + 2x}, \text{ then } \frac{dy}{dx} = (12x^3 + 2)e^{3x^4 + 2x}$$

Exercises:

Differentiate the following with respect to x .

(a) $y = e^{3+2x}$ $f(x) = 3+2x$ $f'(x) = 2$ $\therefore y' = 2e^{3+2x}$

(b) $y = 2e^{-3x^5}$ $f(x) = -3x^5$ $f'(x) = -15x^4$ $\therefore y' = 2 \times (-15x^4) e^{-3x^5}$
 $= -30x^4 e^{-3x^5}$

(c) $y = \frac{e^{2x}}{2}$ $f(x) = 2x$ $f'(x) = 2$ $\therefore y' = \frac{1}{2} \times 2 e^{2x}$
 $= e^{2x}$

(d) $y = 2e^{3-4x^2}$ $f(x) = 3-4x^2$ $f'(x) = -8x$ $\therefore y' = 2 \times (-8x) e^{3-4x^2}$
 $= -16x e^{3-4x^2}$

Exercises:

Differentiate the following with respect to x .

(a) $y = 5e^{\frac{x}{3}}$

$$\begin{aligned} f(x) &= \frac{x}{3} & \therefore y' &= 5 \times \frac{1}{3} e^{\frac{x}{3}} \\ f'(x) &= \frac{1}{3} & &= \frac{5}{3} e^{\frac{x}{3}} = \frac{5e^{\frac{x}{3}}}{3} \end{aligned}$$

(b) $y = 2x^4 + 4e^{2x}$

$$\begin{aligned} f(x) &= 2x & \therefore y' &= 8x^3 + 4 \times 2e^{2x} \\ f'(x) &= 2 & &= 8x^3 + 8e^{2x} \end{aligned}$$

(c) $y = e^{4x}(e^x - e^{-2x})$

expand first

$$\begin{aligned} y &= e^{4x} \times e^x - e^{4x} \times e^{-2x} \\ y &= e^{5x} - e^{2x} \end{aligned}$$
$$\therefore y' = 5e^{5x} - 2e^{2x}$$

Example using product rule: Differentiate $y = x^2 e^{3x}$

$$y' = vu' + uv'$$

$$= e^{3x} \times 2x + x^2 \times 3e^{3x}$$

$$= 2xe^{3x} + 3x^2 e^{3x}$$

$$= xe^{3x}(2 + 3x)$$

$$u = x^2$$

$$u' = 2x$$

$$v = e^{3x}$$

$$v' = 3e^{3x}$$

Example using chain rule: Differentiate $y = (e^{3x^2-1})^4$

$$\frac{dy}{dx} = 4(e^{3x^2-1})^3 \times 6xe^{3x^2-1}$$

$$= 24x(e^{3x^2-1})^4$$

Example using quotient rule: Differentiate $y = \frac{e^x}{2x}$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{2xe^x - 2e^x}{4x^2}$$

$$= \frac{2e^x(x-1)}{4x^2}$$

$$= \frac{e^x(x-1)}{2x^2} \quad \text{or} \quad y' = \frac{xe^x - e^x}{2x^2}$$

$$u = e^x$$

$$u' = e^x$$

$$v = 2x$$

$$v' = 2$$

Exercise:

Differentiate the following functions:

$$y = u \times v \quad y' = u'v + uv'$$

(a) $y = \sqrt{x}e^{2x-1}$

$$u = \sqrt{x} = x^{\frac{1}{2}}$$

$$u' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$v = e^{2x-1}$$

$$v' = 2e^{2x-1}$$

$$\therefore y' = \frac{1}{2\sqrt{x}} \times e^{2x-1} + \sqrt{x} \times 2e^{2x-1}$$

$$= \frac{e^{2x-1}}{2\sqrt{x}} + 2\sqrt{x}e^{2x-1}$$

$$= e^{2x-1} \left(\frac{1}{2\sqrt{x}} + 2\sqrt{x} \right)$$

(b) $y = e^{-x}(2x+x^2)$

$$u = e^{-x} \quad v = 2x+x^2$$

$$u' = -e^{-x} \quad v' = 2+2x$$

$$\therefore y' = -e^{-x} \times (2x+x^2) + e^{-x} \times (2+2x)$$

$$= e^{-x} \left(-(2x+x^2) + (2+2x) \right)$$

$$= e^{-x} \left(-2x - x^2 + 2 + 2x \right)$$

$$= e^{-x} \left(2 - x^2 \right)$$

$$(c) \quad y = \frac{e^x + 1}{3x - 1}$$

$$y = \frac{u}{v}$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$u = e^x + 1 \quad v = 3x - 1$$

$$u' = e^x \quad v' = 3$$

$$y' = \frac{e^x(3x-1) - 3(e^x+1)}{(3x-1)^2} = \frac{3xe^x - e^x - 3e^x + 3}{(3x-1)^2}$$

$$= \frac{3xe^x - 4e^x + 3}{(3x-1)^2}$$

$$(d) \quad y = (5x + e^{x^2})^3$$

$$y' = 3(5x + e^{x^2})^2 \times (5 + 2xe^{x^2})$$

$$y' = 3(5 + 2xe^{x^2})(5x + e^{x^2})$$

Applications of Exponential Differentiation

Exercise:

Given $y = e^{2x^3-1} + e^{2x}$ find $y''(1)$.

$$y' = \underbrace{6x^2}_{\text{Then use product rule}} e^{2x^3-1} + 2e^{2x}$$

$$u = 6x^2 \quad v = e^{2x^3-1}$$

$$u' = 12x \quad v' = 6x^2 e^{2x^3-1}$$

$$y'' = 12x(e^{2x^3-1}) + 6x^2 \times 6x^2 e^{2x^3-1} + 4e^{2x}$$

$$= 12x e^{2x^3-1} + 36x^4 e^{2x^3-1} + 4e^{2x}$$

$$y''(1) = 12 \times 1 e^{2 \times 1^3 - 1} + 36 \times 1^4 e^{2 \times 1^3 - 1} + 4e^{2 \times 1}$$

$$= 12e^1 + 36e^1 + 4e^2 = 48e + 4e^2$$

Example:

Find the equation of the tangent to the curve $y = 2e^x + 1$ at the point where the curve crosses the y -axis. Write the equation in the general form.

Step 1: The curve cuts the y -axis at $x = 0$.

$$\text{If } x = 0 \text{ then } y = 2e^0 + 1 = 3$$

Therefore, the curve passes through the point $(0, 3)$.

$$y = 2e^x + 1$$

$$y' = 2e^x$$

Step 2: If $x = 0$ then $y' = 2e^x = 2e^0 = 2$

Therefore the gradient is $m = 2$

Step 3: For equation of tangent use

$$y - y_1 = m(x - x_1)$$

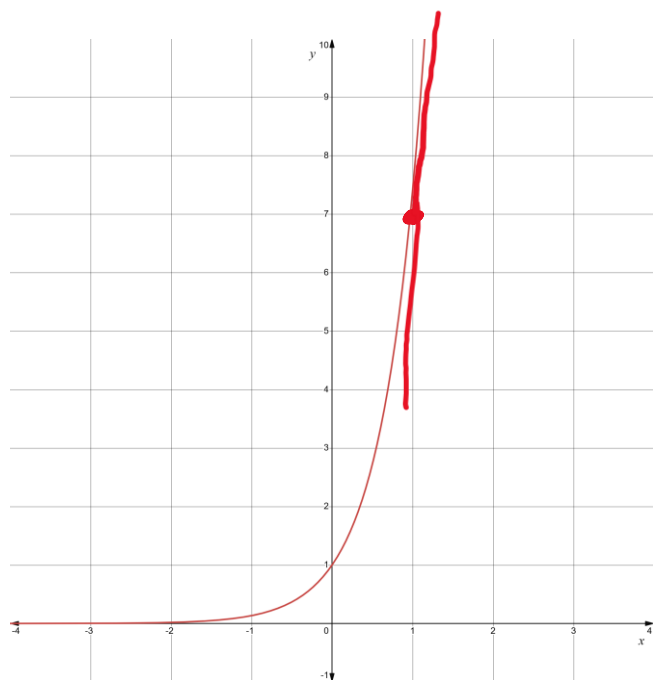
$$y - 3 = 2(x - 0)$$

$$y = 2x + 3$$

$$2x - y + 3 = 0 \quad \text{General Form}$$

Exercise:

Find the equation of the tangent to the curve $y = e^{2x}$ at $(1, e^2)$.



- find gradient m

$$m = y' = 2e^{2x}$$

$$y'(1) = 2e^{2 \times 1} \\ = 2e^2$$

- equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - e^2 = 2e^2(x - 1)$$

$$y - e^2 = 2xe^2 - 2e^2$$

$$y = 2xe^2 - 2e^2 + e^2$$

$$y = 2xe^2 - e^2$$